Q1. (40pts)

Answer both parts below assuming a hash table with size *m=13* with the following hash function:

index = (key mod m)

Show the contents of the hash table after the following operations have been performed. Indicate next to each operation the number of cells examined.

i. Assume that linear probing is used to resolve collisions and that the hash table is initially empty.

	cells
insert 7	1
insert 21	1
insert 33	3
insert 48	2
delete 21	1
search 33	2
search 21	1

0	
1	
2	
3	
4	
5	
6	
7	7
8	~~21~~
9	33
10	48
11	
12	

ii. Assume that quadratic probing is used to resolve collisions and that the hash table is initially empty.

	cells
insert 10	1
insert 36	2
insert 49	3
search 10	1
delete 36	2
insert 24	1
search 23	4

0	
1	49
2	
3	
4	
5	
6	
7	
8	
9	
10	10
11	~~36~~ 24
12	

Q2. (60pts)

A hash table with the hash function h(k)=k mod m uses the double hashing strategy to resolve collisions. The first three keys 193, 297 and 321 produce the hash values 3, 12 and 17, respectively.

a) (15pts) What is the *prime* hash table size?

$$\left.\begin{array}{l} 193 \bmod m = 3 \\ 297 \bmod m = 12 \\ 321 \bmod m = 17 \end{array}\right\} \Rightarrow \begin{array}{l} (193-3) = km; k \in N \\ (297-12) = lm; l \in N \\ (2366-20) = nm; n \in N \end{array} \quad \begin{array}{l} 190 = 2*5*19 \\ \Rightarrow 285 = 3*5*19 \Rightarrow m = 19 \\ 321 = 2^4 * 19 \end{array}$$

b) (20+15 pts) The next key (i.e., the fourth key) that is hashed to the table after three collisions (each collision with a different key) is 326.

1. What is R in the double hash function?
2. Where is 326 hashed in the table?

Hash function using the double hashing strategy $h_{dh}(k,i)$:
$h(k,i)=[h(k)+i\ h_{dh}(k,i)]\ mod\ m$ where $h_{dh}(k,i)=R-(\ k\ mod\ R\ )$

From the question we know that, 326 collides with all three keys in the hash table. Checking to find the same interval between the locations of the hashed keys we find that it is 14 between 3-17-12 (i.e., 31 mod 19=12).
That is,

$h_{dh}(326,1)=R-(\ 326\ mod\ R\ )=14 \Rightarrow [326-(R-14)]=kR,\ k \in N$
$326+14=(k+1)*R; 340=2*2*5*17 \Rightarrow \mathbf{R=17}$ (Answer to 2.b.1)

After the third collision,

$h(326,3)=[3+3*14]\ mod\ 19=7.$

326 gets placed in hash table cell #7. (Answer to 2.b.2)

c) (10pts) Place, in respective order, the keys 107 and 127. In case of collisions use the double hashing strategy with the above R you found. Specify the number of possible collisions during hashing both keys!

$h(107)=107\ mod\ 19=12 \Rightarrow$ collision w/297; 17-[107 mod 17]=12
$h(107,1)=[12+1*12]\ mod\ 19=5 \Rightarrow$ **107 is placed in #5 with 1 collision!**
$h(127)=127\ mod\ 19=13 \Rightarrow$ **127 is placed in #13 with no collisions!**